

Lab - Shared Access Signatures (SAS) - At the Blob Level

To Begin with the Lab

Summary

This lab demonstrates how to create a **Shared Access Signature (SAS)** for an individual blob in Azure Storage. A Blob SAS provides secure, temporary, and limited access to a specific blob without exposing the storage account access keys.

Understanding

A **Shared Access Signature (SAS)** is a secure URL that grants limited access to Azure Storage resources. Unlike **Access Keys**, which provide full access to the entire storage account, a **Blob SAS** allows access only to a specific blob with defined permissions (such as Read, Write, or Delete) and only during a specified time period.

When generating a Blob SAS, Azure signs the URL using the storage account's access key. The generated URL contains a security token that includes the permissions, start time, expiry time, and signature. Once the SAS expires, the URL automatically becomes invalid.

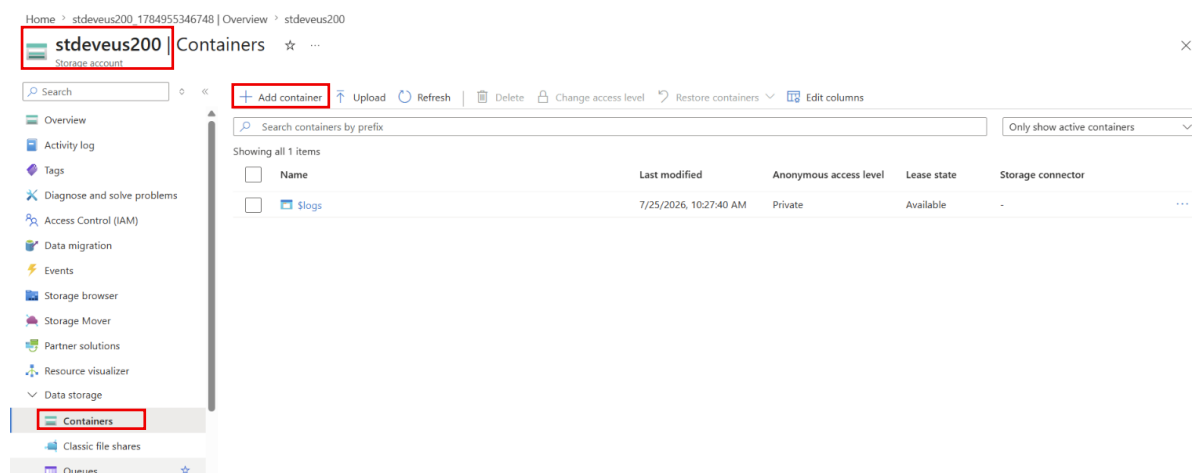
Example:

Suppose you upload a training video to Blob Storage and want users to access it for only **7 days**. Instead of sharing your storage account credentials, you generate a **Blob SAS** with **Read** permission and a **7-day expiry**. Users can view the video only until the expiration time.

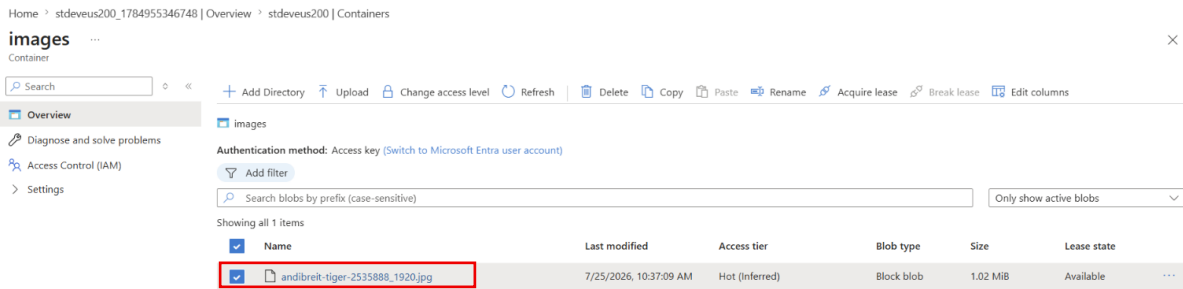
Lab Steps

Step 1: Open the Blob Container

- Open the Azure Portal.
- Navigate to your **Storage Account**.
- Select **Containers**.
- Click on the Add container (**images**).

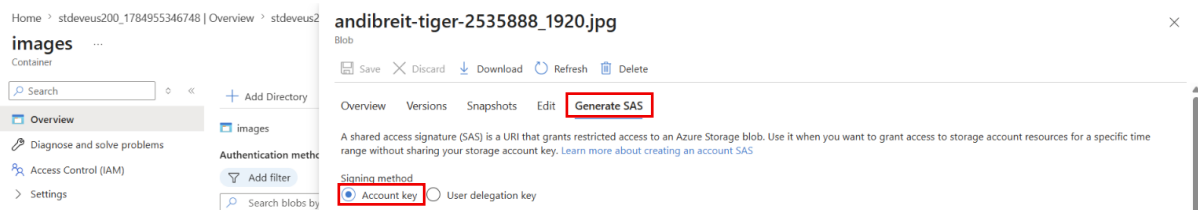


- Open the container (**images**).
- Select the any blob (image or file) for which you want to generate a SAS.



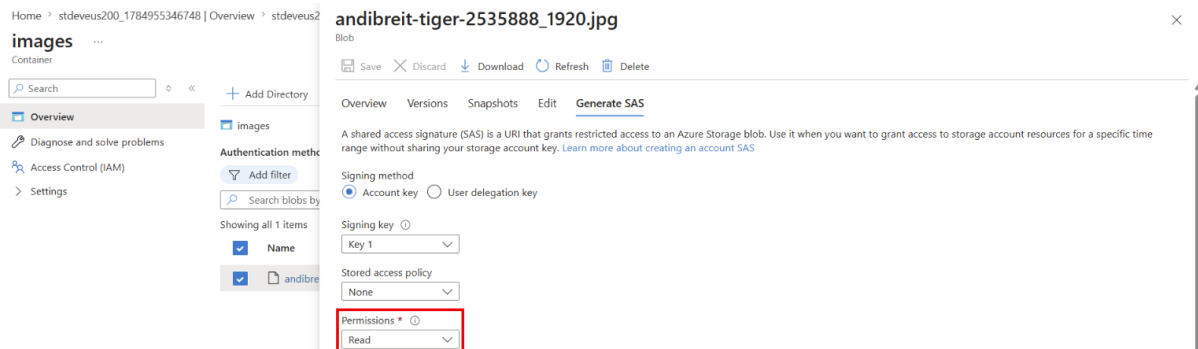
Step 2: Generate a Shared Access Signature

- Click **Generate SAS** for the selected blob.
- Choose the **Signing Method** as **Account Key**.
- Azure uses the storage account key to create the SAS token.
- Click on the **Generate SAS token and URL**.



Step 3: Configure Blob Permissions

- Select the required permissions.
- For this lab, enable only:
 - Read**
- Leave other permissions disabled unless needed.



Step 4: Set the Validity Period

- Specify the **Start Date and Time**.
- Specify the **Expiry Date and Time**.
- The SAS URL will remain valid only during this time window.

Start and expiry date/time ⓘ

Start

07/25/2026 10:24:33 AM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Expiry

07/25/2026 6:39:33 PM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Step 5: Generate the SAS Token

- Click **Generate SAS Token and URL**.
- Azure generates:

- Blob SAS Token
- Blob SAS URL

andibreit-tiger-2535888_1920.jpg

Blob

Save Discard Download Refresh Delete

Signing method

☒ Account key ☐ User delegation key

Signing key ⓘ

Key 1

Stored access policy

None

Permissions * ⓘ

Read

Start and expiry date/time ⓘ

Start

07/25/2026 10:24:33 AM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Expiry

07/25/2026 6:39:33 PM

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Allowed IP addresses ⓘ

for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ⓘ

☒ HTTPS only ☐ HTTPS and HTTP

Generate SAS token and URL

Step 6: Copy the Blob SAS URL

- Copy the generated **Blob SAS URL**.

andibreit-tiger-2535888_1920.jpg

Blob

Save Discard Download Refresh Delete

Key 1

Stored access policy

None

Permissions * ⓘ

Read

Start and expiry date/time ⓘ

Start

07/25/2026 10:24:33 AM

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Expiry

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Allowed IP addresses ⓘ

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Allowed protocols ⓘ

☒ HTTPS only ☐ HTTPS and HTTP

Generate SAS token and URL

Blob SAS token ⓘ

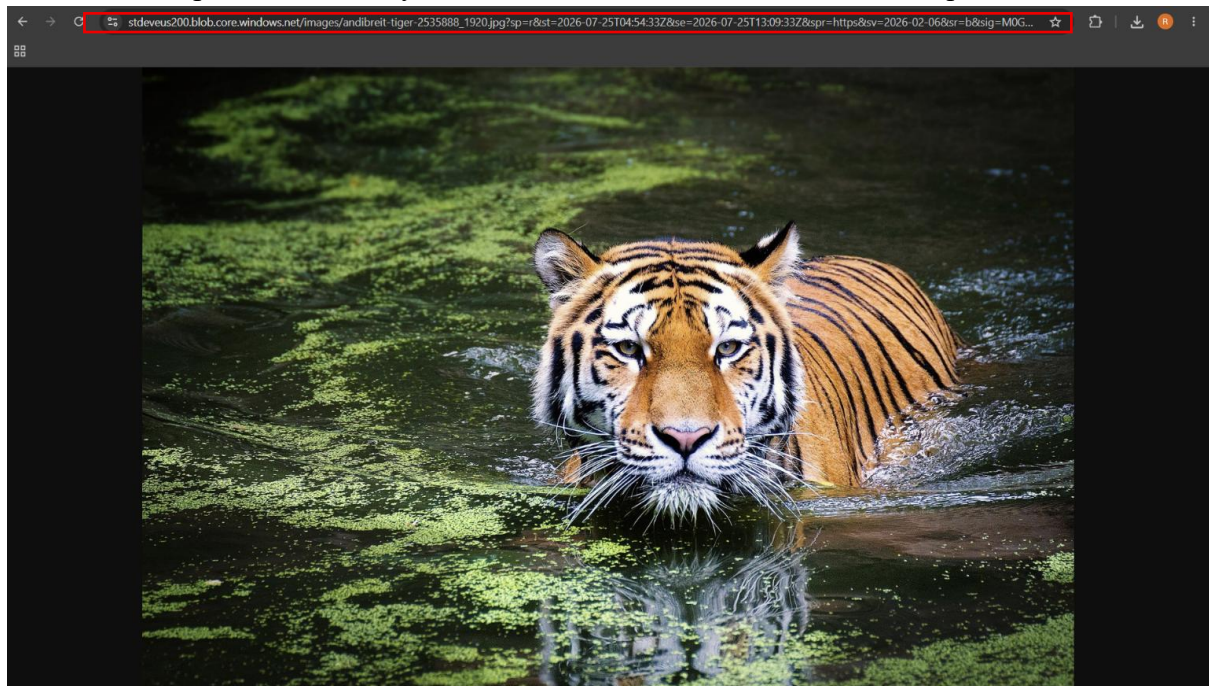
sp=r&st=2026-07-25T04:54:33Z&se=2026-07-25T13:09:33Z&spr=https&sv=2026-02-06&sr=b&sig=M0GEpoRnpXU8zxz%2F%2BwYBjht5lmiSBb9GWtoQbKUj6AA%3D

Blob SAS URL

https://stdeveus200.blob.core.windows.net/images/andibreit-tiger-2535888_1920.jpg?sp=r&st=2026-07-25T04:54:33Z&se=2026-07-25T13:09:33Z&spr=https&sv=2026-02-06&sr=b&sig=M0GEpoRnpXU8zxz%2F%2BwYBjht5lmiSBb9GWtoQbKUj6AA%3D

- Open a new browser tab.
- Paste the URL into the address bar.

- The blob opens successfully because the SAS token authorizes the request.



Step 7: Understand the SAS URL

The generated URL contains:

- Storage Account Name
- Blob Service Endpoint
- Container Name
- Blob Name
- SAS Token containing:
 - Permissions
 - Start Time
 - Expiry Time
 - Signature

`https://stdeveus200.blob.core.windows.net/images/andibreit-tiger-2535888_1920.jpg?sp=r&st=2026-07-25T04:54:33Z&se=2026-07-25T13:09:33Z&spr=https&sv=2026-02-06&sr=b&sig=M0GEpoRnpXU8zxz%2F%2BwYBjht5ImiSBb9GWtoQbKUj6AA%3D`

Step 8: Verify Expiration

- Access the blob before the expiry time.
- After the expiry time, open the same SAS URL again.
- Azure denies access because the SAS token has expired.

